

Button and Fold a Dress Shirt SOP

One episode buttons one long sleeved dress shirt and folds it. The shirt starts crumpled in the **shirt spot**, which changes with the config. The gripper on the shirt's side takes it and lays it in the middle. Both grippers spread it face up, collar toward the far side and sleeves out to the sides. The front buttons are closed one at a time from the bottom up, and the placket is smoothed. The shirt is flipped onto its front by the shoulders so the collar lies toward the front edge, and it is straightened. The right sleeve and then the left sleeve are folded onto the back. The shirt is turned 90 degrees clockwise, the near side and then the far side are folded in, the shirt is turned 45 degrees clockwise, and the near end is folded up and the far end folded down over it. The right gripper turns the finished packet round and lays it in the **finished spot**, back-right or right-center depending on the config, collar facing the far side.

The shirt closes with normal sewn **buttons** through **buttonholes**, not press snaps. A button is closed by laying the buttonhole over it and drawing the cloth across until the button comes out through the hole.

The order never changes: move, spread, button, smooth, flip, straighten, sleeves, turn, sides, turn, end folds, place. Nothing is folded before every button is closed and the shirt is face down. The sides are not folded before the first turn, and the ends are not folded before the second turn.

Every fold is made on the desk in the **work zone**. A gripper lifts the cloth it folds and lays it down. Nothing is pushed across the desk to fold it. Any grasp may be retried: if a gripper slips off the cloth, it opens, backs off, and takes the same grasp again.

Adjustments are allowed on every step. Either gripper or both may pull a corner straight, push a sleeve, an edge, or a fold back into line, flatten a bunch, or set the shirt or the packet square, at any time. An adjustment is a fix, not a step.

The right gripper does the primary work: it lays each buttonhole over its button and draws the button through, smooths the placket, brings the right sleeve to the middle of the back and finishes laying the left sleeve, and turns the finished packet and lays it in the finished spot. The left gripper supports: it holds each button while it is worked, holds the middle of the hem while the placket is smoothed, and brings the left sleeve to the middle of the back and finishes laying the right sleeve. Either gripper may hold the button. The **pick gripper** moves the shirt from the shirt spot to the middle. Both grippers work together, one on each side, to spread, flip, straighten, turn, and fold the shirt.

The desk is set up in one of twelve ways. Two things move: the shirt spot, where the shirt starts, and the finished spot, where the packet ends. The work zone is in the same place in all twelve.

- **Config BL-BR:** the shirt starts in the back-left corner. The packet ends in the back-right corner.

- **Config BL-RC:** the shirt starts in the back-left corner. The packet ends in the right-center.
- **Config BC-BR:** the shirt starts in the back-center. The packet ends in the back-right corner.
- **Config BC-RC:** the shirt starts in the back-center. The packet ends in the right-center.
- **Config BR-BR:** the shirt starts in the back-right corner. The packet ends in the back-right corner.
- **Config BR-RC:** the shirt starts in the back-right corner. The packet ends in the right-center.
- **Config LC-BR:** the shirt starts in the left-center. The packet ends in the back-right corner.
- **Config LC-RC:** the shirt starts in the left-center. The packet ends in the right-center.
- **Config FC-BR:** the shirt starts in the front-center. The packet ends in the back-right corner.
- **Config FC-RC:** the shirt starts in the front-center. The packet ends in the right-center.
- **Config RC-BR:** the shirt starts in the right-center. The packet ends in the back-right corner.
- **Config RC-RC:** the shirt starts in the right-center. The packet ends in the right-center.

In Config BR-BR and Config RC-RC the packet ends in the same spot the shirt started in. The shirt leaves that spot in Step 1, so it is bare again by Step 12.

Where a step depends on the setup it says so on an **IF** line: look at the desk and follow the line that matches.

What stays constant across all sessions:

- **One config per episode:** chosen before recording and never changed mid-episode.
- **Same-side rule:** the pick gripper takes the shirt: the left gripper when the shirt starts back-left, back-center, left-center, or front-center, and the right gripper when it starts back-right or right-center. No arm reaches across the desk for the shirt.
- **Finished spot:** the right gripper always turns the packet and lays it down, at the back-right or the right-center.
- **Fixed roles:** Steps 2 to 11 are the same in all twelve configs.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the shirt spot, the finished spot, and the whole work zone, with clear desk all round the spread shirt: collar, hem, both sleeves, and both cuffs in frame.

3. The environment camera also shows the space above the work zone, so the shirt stays in frame while it hangs from both shoulders during the spread and the flip.
4. The overhead view shows the front placket end to end, with every button and buttonhole readable.
5. Both arms are at home with grippers open.
6. The desk is clear and bare apart from the shirt.
7. The left gripper reaches the back-left, back-center, left-center, and front-center shirt spots, the left half of the work zone, the left cuff, the middle of the hem, and every button on the placket without the arm leaning out or reaching a joint limit.
8. The right gripper reaches the back-right and right-center spots, the right half of the work zone, the right cuff, and every buttonhole on the placket without the arm leaning out or reaching a joint limit.
9. The right gripper can hold the finished packet and turn it until the collar faces the far side without reaching a joint limit.
10. Both grippers together can lift the shirt by its two shoulder points until the hem hangs clear of the desk, and can carry that hold through the flip without either arm reaching a joint limit.
11. Both grippers can hold the two turn corners and carry them round 90 degrees clockwise for the first turn and 45 degrees clockwise for the second, without the arms striking each other or reaching a joint limit.

Materials checklist

1. One **long sleeved dress shirt** lying crumpled in a loose heap in the shirt spot for the episode's config, inside the camera frame, with clear desk between it and the work zone. Every other spot is bare.
2. The shirt closes with normal sewn **buttons** down the front, not press snaps. As seen from the front edge with the shirt face up, the buttons are on the left and the buttonholes on the right.
3. Every front button is undone at the start. The collar band button and both cuff buttons are also undone, and they stay undone for the whole episode.
4. Every front button is sewn on firm, and every buttonhole passes its button on one draw.
5. The shirt is dry and free of stains, tears, and open seams. It holds a fold rather than springing open.
6. The finished spot for the config is bare, unless the shirt starts there (Config BR-BR and Config RC-RC).
7. The desk surface is dry, clean, and free of anything the cloth can catch on.

Workspace layout

- **Shirt spot:** where the shirt starts, crumpled. It is one of six places, set by the config:

- **back-left:** the back-left corner of the desk.
- **back-center:** behind the work zone, in the middle of the back edge.
- **back-right:** the back-right corner of the desk.
- **left-center:** the left side of the desk, beside the middle of the work zone.
- **front-center:** between the work zone and the front edge, in the middle.
- **right-center:** the right side of the desk, beside the middle of the work zone.

Only the pick gripper takes it.

- **Work zone:** the center of the desk, reachable by both arms. It holds the one shirt. Every button, every fold, and both turns happen here.
- **Finished spot:** where the packet ends, set by the config: the back-right corner, or the right-center, on the right side of the desk beside the middle of the work zone. Only the right gripper lays the finished packet here, collar facing the far side.
- **Left half of the work zone:** the left gripper's side. **Right half of the work zone:** the right gripper's side. Each arm keeps to its own half plus the shared middle.
- **Space above the work zone:** the clear air where the shirt hangs from both shoulders during the spread and the flip. Nothing else stands on the desk under it.

Arm assignments

- **Right gripper:** works. It lays each buttonhole over its button and draws the button through, smooths the placket, brings the right sleeve to the middle of the back, finishes laying the left sleeve, and turns the finished packet and lays it in the finished spot. It is also the pick gripper when the shirt starts back-right or right-center.
- **Left gripper:** supports. It is the pick gripper when the shirt starts back-left, back-center, left-center, or front-center. It holds each button while it is worked, holds the middle of the hem while the placket is smoothed, brings the left sleeve to the middle of the back, and finishes laying the right sleeve. Either gripper may hold the button.
- **Both grippers:** spread the shirt, flip it, straighten it, turn it, and make the side folds and the two end folds, one on each side.

Vocabulary

- **Left and right:** as seen from the front edge, at that moment. The **left sleeve** is the sleeve lying on the left of the desk, whichever way the shirt faces.
- **Near and far:** near is toward the front edge of the desk, far is toward the back.
- **Clockwise:** the way a clock's hands turn, as seen from above.
- **Face up:** the front of the shirt, with the buttons, shows. **Face down:** the front lies on the desk and the back shows.

- **Crumpled:** the shirt lies in a loose heap with no side laid out flat.
- **Shoulder point:** the top corner of the shirt where the shoulder seam meets the sleeve. There is one on the left and one on the right.
- **Placket:** the doubled strip down the front of the shirt. The **button placket** carries the buttons and lies on the left when the shirt is face up. The **hole placket** carries the buttonholes and lies on the right.
- **Bottom button:** the button closest to the hem. **Top button:** the front button closest to the collar. Buttons are worked from the bottom button up to the top button. The collar band button is not one of them.
- **Matching buttonhole:** the buttonhole in the same place, counting from the hem, as the button being worked. The bottom button goes through the bottom buttonhole, the next button through the next hole, and so on.
- **Hold the button:** a gripper pinches the button and keeps hold of it while the buttonhole is worked over it, until the button is through.
- **Buttoned:** the whole button sits flat on top of the hole placket with its whole rim clear of the buttonhole, and it stays there when both grippers lift away.
- **Smooth:** a closed gripper pressed down on the cloth and drawn along in one run, end to end, without lifting off.
- **Straight:** the shirt lies flat with no fold or bunch in the body, both sleeves lie out flat to the sides, and the collar and hem lie square across the desk.
- **Fold onto the back:** the gripper on the sleeve's side pinches the cuff, lifts the sleeve, carries it over the back, and lays it down at the middle of the back. The other gripper takes over and lays the rest of the sleeve down flat.
- **Turn corners:** the two corners a turn step names, as the shirt lies at that moment. In Step 8 they are the far left corner (left gripper) and the near right corner (right gripper). In Step 10 they are the far left corner (left gripper) and the far right corner (right gripper).
- **Turn:** the shirt turns clockwise round its middle on the desk by the degrees the step gives. The two grippers move together round the middle, and the rest of the shirt slides flat on the desk. The wrists do not twist.
- **90 degrees:** each side of the shirt ends where the next side was. A collar lying toward the front edge ends on the left.
- **45 degrees:** half of 90 degrees. A strip lying straight across the desk ends lying corner to corner, one end toward the far left and the other toward the near right.
- **Fold edge:** the closed folded side a fold makes.
- **Collar end:** the end of the shirt where the collar is.
- **Packet:** the shirt once both end folds are made: one folded block with the collar on top.
- **Pick gripper:** the gripper that takes the shirt from the shirt spot and lays it in the middle: the gripper on the shirt's side. The other gripper waits clear of the shirt in Step 1.

- **Adjustment:** a small fix to the shirt or the packet: pulling a corner straight, pushing a sleeve, an edge, or a fold back into line, flattening a bunch, or setting the shirt or the packet square. Either gripper or both may make it, on any step. It does not change the step's arm roles.
- **Settled:** the cloth stays where it was laid when the gripper lifts away, with nothing sliding or springing open.
- **Let go:** a gripper opens only once the cloth it holds is resting on the desk.

Steps

Run Steps 1 to 12 once, in order, on the one shirt. Step 13 ends the episode.

Adjustments are allowed on every step, by either gripper or both. Where a Check names a gripper for a fix, that fix is an adjustment too, and either gripper or both may make it.

Only Steps 1 and 12 depend on the config. In Step 1 the IF lines say which gripper is the pick gripper. In Step 12 they say which finished spot the packet goes to. Every other step is the same in all twelve configs.

Step 1: Move the shirt to the middle

Goal: the crumpled shirt lying in the middle of the work zone.

- **IF the shirt starts back-left, back-center, left-center, or front-center:** the **left gripper** is the pick gripper and the **right gripper** waits clear of the shirt.
- **IF the shirt starts back-right or right-center:** the **right gripper** is the pick gripper and the **left gripper** waits clear of the shirt.

Then, in all twelve:

- The **pick gripper** pinches the top of the crumpled shirt in the shirt spot.
- The **pick gripper** lifts the shirt clear of the desk, carries it to the middle of the work zone, and lays it down. Let go once it is resting.

Check: the whole shirt lies in the work zone and the shirt spot is bare. If part of the shirt still lies in the shirt spot or hangs off the work zone, the **pick gripper** pinches the shirt again, lifts it, and lays it down in the middle.

Step 2: Spread the shirt face up

Goal: the shirt lying **straight** and face up in the work zone, collar toward the far side, hem toward the front edge, sleeves out to the sides.

Run 2.1, then 2.2, then 2.3.

2.1 Lift the shirt by the shoulders

- The **left gripper** pinches the left **shoulder point**.
- The **right gripper** pinches the right shoulder point.
- **Both grippers** lift together until the hem hangs clear of the desk and the buttons face the front edge.
- If the back faces the front edge, the **left gripper** and the **right gripper** lay the shirt back down in the middle, and 2.1 runs again.

2.2 Lay the shirt down face up

- **Both grippers** bring the shirt down until the hem touches the desk near the front edge.
- **Both grippers** carry the shoulders toward the far side and down, so the shirt lays itself down face up with the collar toward the far side. Let go once it is resting.

2.3 Straighten the shirt face up

- **Both grippers** spread the shirt out, each working a different side of it, until the shirt lies well flat. There is no set order and no set grip.

Check: the shirt is face up and straight, the collar is toward the far side, both sleeves lie out to the sides, and every button shows beside the hole placket. If the shirt landed face down or the collar lies toward the front edge, run 2.1 again.

Step 3: Button the front from the bottom up

Goal: every front button closed, from the bottom button to the top button.

Run 3.1, then 3.2, then 3.3 for the bottom button. Then run them again for each button above it, one at a time, until the top button is done.

3.1 Hold the button

- The **left gripper** pinches the button being worked and **holds the button**.

3.2 Lay the buttonhole over the button

- The **right gripper** pinches the outer edge of the hole placket in line with the **matching buttonhole**, lifts it a little off the cloth below, carries it across to the left, and lays the buttonhole directly on top of the button.

3.3 Draw the button through

- The **right gripper** pinches the hole placket just past the far lip of the buttonhole and draws it straight across to the left, until the whole button comes out through the hole and lies flat on top of the hole placket.
- A gripper holds the button while the other gripper draws. Either gripper may hold the button.
- **Both grippers** may make any adjustment that helps the button go into the hole and through it: pinch again, push, or pull the cloth, or swap which gripper holds the button.

- Once the button is through, the **left gripper** and the **right gripper** open and lift away.

Check: the button is **buttoned**, it went through its matching buttonhole, and every button below it is still closed. If the button slips back out or only part of its rim came through, the **right gripper** lifts the hole placket back off the button and 3.2 runs again. If the button went through the wrong hole, the **right gripper** and the **left gripper** undo it and 3.1 runs again for that button.

Step 4: Smooth the placket

Goal: the closed placket lying flat and straight from the collar to the hem.

- The **left gripper** pinches the middle of the hem, at the bottom of the placket, and holds it there until the smooth is done.
- The **right gripper**, closed, **smooths** up the placket from the hem to the collar in one run, away from the left gripper's hold.
- The **right gripper** lifts away first, then the **left gripper**.

Check: the placket runs straight from the collar to the hem, lies flat with no bunch along it, and every button is still closed. If the placket is still bunched, run Step 4 again. If a button came open, close it again as in Step 3 and smooth again.

Step 5: Flip the shirt face down by the shoulders

Goal: the shirt lying face down in the work zone with the collar toward the front edge and the hem toward the far side.

- The **left gripper** pinches the left shoulder point.
- The **right gripper** pinches the right shoulder point.
- **Both grippers** lift together until the hem hangs clear of the desk.
- **Both grippers** carry the shoulders toward the front edge and down, so the shirt lays itself over onto its front with the collar nearest the front edge. Let go once the whole shirt is resting.

The arms carry the shoulders over. The wrists do not turn the shirt.

Check: the shirt is face down in the work zone, the collar is toward the front edge, and every button is still closed. If the shirt landed face up or the collar lies toward the far side, the **left gripper** and the **right gripper** take the shoulder points again and run Step 5 again.

Step 6: Straighten the shirt face down

Goal: the shirt lying **straight** and face down, collar toward the front edge, sleeves out to the sides.

- The **left gripper** pinches the left cuff and lays the left sleeve out flat to the left.

- The **right gripper** pinches the right cuff and lays the right sleeve out flat to the right.
- If the body is bunched, the **left gripper** pinches the left hem corner and the **right gripper** pinches the right hem corner, and both pull a little toward the far side until it lies flat.

Check: the shirt is face down and straight, no sleeve lies under the body, and both cuffs lie clear of the body. If a sleeve is still under the body or doubled back, that side's gripper lays it out again.

Step 7: Fold both sleeves onto the back

Goal: both sleeves lying flat on the back of the shirt, right sleeve first.

Run 7.1, then 7.2.

7.1 Fold the right sleeve

- The **right gripper** pinches the right cuff, lifts the sleeve, carries it over the back, and lays the cuff down at the middle of the back. Let go once it is resting.
- The **left gripper** takes over: it pinches the cuff and lays the rest of the sleeve down flat, running from the shoulder toward the hem. Let go once it is resting.

7.2 Fold the left sleeve

- The **left gripper** pinches the left cuff, lifts the sleeve, carries it over the back, and lays the cuff down at the middle of the back. Let go once it is resting.
- The **right gripper** takes over: it pinches the cuff and lays the rest of the sleeve down flat, running from the shoulder toward the hem. Let go once it is resting.

Check: both sleeves lie flat on the back, each cuff points toward the hem, and no part of a sleeve hangs off the side of the body. If a sleeve landed bunched or off the body, a gripper pinches its cuff and lays it down again.

Step 8: Turn the shirt 90 degrees clockwise

Goal: the shirt lying across the work zone with the collar end on the left and the hem on the right, both sleeves still on the back.

- The **left gripper** pinches the far left **turn corner**.
- The **right gripper** pinches the near right turn corner.
- **Both grippers** lift their corners a little and move together round the middle of the shirt, the **left gripper** toward the right along the far side and the **right gripper** toward the left along the near side, until the shirt has **turned** 90 degrees clockwise.
- Let go once the shirt is resting.

Check: the collar end is on the left, the hem is on the right, the shirt lies in the work zone, and both sleeves are still on the back. If the shirt turned too far or not far enough, the **left gripper** and the **right gripper** take the turn corners again and finish the turn.

Step 9: Fold the near side and the far side in

Goal: both sides folded in on straight fold edges, so the shirt is one long strip about as wide as the collar.

Run 9.1, then 9.2.

9.1 Fold the near side

- The **left gripper** pinches the near side of the shirt close to the collar end.
- The **right gripper** pinches the near side close to the hem.
- **Both grippers** lift the near side together, carry it toward the far side, and lay it down so the new fold edge lines up with the near end of the collar. Let go once it is resting.

9.2 Fold the far side

- The **left gripper** pinches the far side of the shirt close to the collar end.
- The **right gripper** pinches the far side close to the hem.
- **Both grippers** lift the far side together, carry it toward the front edge, and lay it down so the new fold edge lines up with the far end of the collar. Let go once it is resting.

Check: both fold edges run straight from the collar end to the hem, the strip is about as wide as the collar, and no sleeve or cuff shows past a fold edge. If a fold edge landed crooked, the **left gripper** and the **right gripper** lift that side and lay it down again.

Step 10: Turn the shirt 45 degrees clockwise

Goal: the strip lying corner to corner across the work zone, collar end toward the far left and hem toward the near right.

- The **left gripper** pinches the far left corner.
- The **right gripper** pinches the far right corner.
- **Both grippers** lift their corners a little and move together round the middle of the strip, the **left gripper** toward the far side and the **right gripper** toward the front edge, until the strip has turned 45 degrees clockwise.
- Let go once the strip is resting.

Check: the strip lies corner to corner, the collar end is toward the far left, the hem is toward the near right, and both side folds are still shut. If the strip turned too far or not far enough, the **left gripper** and the **right gripper** take the turn corners again and finish the turn. If a side fold sprang open, the **left gripper** and the **right gripper** lift that side and lay it down again as in Step 9.

Step 11: Fold the near end up and the far end down

Goal: the strip folded in three, near end up first, then far end down over it, so the collar lies on top at the near right end.

Run 11.1, then 11.2.

11.1 Fold the near end up

- The **left gripper** pinches the left corner of the hem end.
- The **right gripper** pinches the right corner of the hem end.
- **Both grippers** lift the hem end together, carry it up along the strip toward the collar end, and lay it down so it covers about the near third of the strip. Let go once it is resting.

11.2 Fold the far end down

- The **left gripper** pinches the left corner of the collar end.
- The **right gripper** pinches the right corner of the collar end.
- **Both grippers** lift the collar end together, carry it down along the strip toward the near fold edge, and lay it down on top so it lines up with the near fold edge. Let go once it is resting.

Check: the shirt is one **packet** in three even layers, the collar end lies on top lined up with the near fold edge, the collar shows face up at the near right end, and no sleeve or cuff shows past an edge. If an end landed short, too far, or crooked, the **left gripper** and the **right gripper** lift that end and lay it down again.

Step 12: Turn the packet and place it in the finished spot

Goal: the packet lying in the finished spot, collar facing the far side.

- **IF the finished spot is back-right:** the packet goes to the back-right corner.
- **IF the finished spot is right-center:** the packet goes to the right side, beside the middle of the work zone.

Then, in all twelve:

- The **right gripper** pinches the right fold edge of the packet at its middle, taking every layer.
- The **right gripper** lifts the packet a little off the desk and turns it until the collar faces the far side.
- The **right gripper** carries the packet to the finished spot and lays it down. Let go once it is resting.
- The **left gripper** stays clear.

Check: the packet lies in the finished spot, the collar faces the far side, and no layer sprang open. If a layer sprang open, the **right gripper** lays the packet down, and the end folds are made again as in Step 11 before the packet is placed again. If the collar does not face the far side, the **right gripper** lifts the packet and turns it again.

Step 13: End the episode

Goal: the packet in the finished spot and both arms home.

- Look once at the packet: every front button closed, both sleeves inside, sides and ends folded in, collar on top facing the far side, and nothing else on the desk.
- Return both arms home with grippers open.
- Stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the folded shirt out of the finished spot and open it out flat.
2. Undo every front button. Check every button is still sewn on firm and every buttonhole still passes its button on one draw.
3. Replace any shirt that is torn, stained, damp, has an open seam, or has a loose or missing button. Vary which shirt is used between episodes rather than running the same one every time.
4. Choose the next episode's config from the twelve. Crumple the shirt into a loose heap and lay it in that config's shirt spot. Do not stage it the same way every time, and vary the config between episodes.
5. Wipe the desk dry and clear, and leave it bare apart from the shirt.
6. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Config misaligned

- **Visible cue:** what happens does not match the config on the desk: the shirt is not in the shirt spot for the config; the gripper on the other side takes the shirt, reaching across the desk; the packet goes to the other finished spot; or the wrong IF line is followed.

- **SOP rule broken:** Steps 1 and 12, the pick gripper takes the shirt: the left gripper when it starts back-left, back-center, left-center, or front-center, the right gripper when it starts back-right or right-center, and the other gripper waits clear of the shirt. The packet goes to the finished spot the config sets.
- **Coaching note:** look at where the shirt is before you move. The gripper on its side takes it.

Violation: Shirt brought in with a button closed

- **Visible cue:** at the start of the episode the shirt in the shirt spot has a front button, the collar band button, or a cuff button already closed.
- **SOP rule broken:** Steps 1 and 3, the shirt comes into the station with every button undone, and the front buttons are closed only in Step 3.
- **Coaching note:** check every button is open before the shirt goes in the shirt spot.

Violation: Shirt dragged or dropped on the way to the middle

- **Visible cue:** the pick gripper pushes or drags the shirt across the desk from the shirt spot instead of lifting it, or opens above the desk so the shirt drops, or part of the shirt is left in the shirt spot.
- **SOP rule broken:** Step 1, the pick gripper lifts the shirt clear, carries it to the middle of the work zone, and lays it down.
- **Coaching note:** pick it up, carry it, put it down.

Violation: Buttoning started before the shirt is spread

- **Visible cue:** a button is worked while the shirt is still crumpled, face down, turned with the collar toward the front edge, with a sleeve under the body, or with the hole placket lying over the buttons.
- **SOP rule broken:** Step 2, the shirt is spread face up and straight, collar toward the far side, sleeves out to the sides, and every button showing before buttoning starts.
- **Coaching note:** spread it first. You cannot button what you cannot see.

Violation: Shirt spread without lifting it by the shoulders

- **Visible cue:** the shirt is pulled open on the desk by its corners, sleeves, or hem instead of being lifted by both shoulder points and laid down face up, or it is laid down by carrying the shoulders toward the front edge.
- **SOP rule broken:** Step 2, both grippers lift the shirt by the two shoulder points and carry the shoulders toward the far side and down to lay it face up.
- **Coaching note:** both shoulders, lift, then lay it down away from you.

Violation: Buttons worked out of order

- **Visible cue:** the first button worked is not the bottom button, a button is closed while a button below it is still open, or the placket is worked from the top down.

- **SOP rule broken:** Step 3, start at the bottom button and work up, one button at a time, to the top button.
- **Coaching note:** start at the hem and count up.

Violation: Button put through the wrong hole

- **Visible cue:** a button comes out through a buttonhole that is not its matching buttonhole, so the two plackets sit out of line and a buttonhole or button is left over at one end.
- **SOP rule broken:** Step 3, each button goes through its matching buttonhole, counting from the hem.
- **Coaching note:** bottom hole to bottom button. Line them up before you draw.

Violation: Moved on from a button that is not through

- **Visible cue:** the episode goes on to the next button or to the smooth while a button is only part way through its hole, slipped back out, or was never worked.
- **SOP rule broken:** Step 3, a button is buttoned, with its whole rim clear of the hole, before the next one is started, and every front button is closed before Step 4.
- **Coaching note:** watch it stay when the grippers lift away. If it slips, run 3.2 again.

Violation: Button worked without holding it

- **Visible cue:** a gripper lays or draws the buttonhole while no gripper is holding the button, or the holding gripper lets go before the button is through.
- **SOP rule broken:** Step 3, a gripper holds the button until the button is through.
- **Coaching note:** hold first, let go last. The button runs away without the hold.

Violation: Placket not smoothed

- **Visible cue:** the shirt goes to the flip with no smooth along the placket, or the right gripper dabs, taps, or smooths from the collar down instead of one run from the hem to the collar, or smooths while the left gripper is not holding the middle of the hem.
- **SOP rule broken:** Step 4, the left gripper holds the middle of the hem while the right gripper, closed, smooths up the placket from the hem to the collar in one run.
- **Coaching note:** hold the middle of the hem, one run up to the collar.

Violation: Shirt flipped by the wrong hold

- **Visible cue:** the shirt is turned over by its hem, collar, a sleeve, or one shoulder, or it is rolled or pushed over on the desk without being lifted, or a wrist twists to turn it.
- **SOP rule broken:** Step 5, both grippers take the two shoulder points, lift until the hem hangs clear, and carry the shoulders toward the front edge and down.
- **Coaching note:** both shoulders, lift, and lay it over toward you. The arms turn it, not the wrists.

Violation: Shirt landed the wrong way after the flip

- **Visible cue:** after the flip the episode goes on with the shirt still face up, or face down with the collar toward the far side.
- **SOP rule broken:** Step 5, the shirt lies face down with the collar toward the front edge.
- **Coaching note:** collar comes to you, buttons go down.

Violation: Sleeves folded before the shirt is straightened

- **Visible cue:** a sleeve is folded onto the back while the other sleeve still lies under the body or doubled back, or while the body is still bunched after the flip.
- **SOP rule broken:** Step 6, both sleeves are laid out flat to the sides and the body lies flat before the sleeves are folded.
- **Coaching note:** lay it out straight after every flip. Sleeves never land right.

Violation: Sleeves folded out of order

- **Visible cue:** the left sleeve is folded onto the back before the right sleeve.
- **SOP rule broken:** Step 7, the right sleeve is folded first, then the left sleeve.
- **Coaching note:** right sleeve, then left sleeve.

Violation: Sleeve not laid flat on the back

- **Visible cue:** a sleeve is left bunched, twisted, hanging off the side of the body, or with the cuff pointing away from the hem, and the episode goes on to the turn.
- **SOP rule broken:** Step 7, each sleeve lies flat on the back, running from the shoulder toward the hem.
- **Coaching note:** lay the whole sleeve down flat, cuff toward the hem.

Violation: Turn made the wrong way or by the wrong hold

- **Visible cue:** the shirt is turned counterclockwise, turned by corners other than the far left and near right in Step 8 or the far left and far right in Step 10, turned by one gripper alone, spun by a wrist twist, turned 45 degrees in Step 8 or 90 degrees in Step 10, or left turned too far or not far enough.
- **SOP rule broken:** Steps 8 and 10, the left gripper takes the far left corner, the right gripper takes the near right corner in Step 8 and the far right corner in Step 10, and both move round the middle until the shirt has turned 90 degrees clockwise in Step 8 and 45 degrees clockwise in Step 10.
- **Coaching note:** take the corners the step names, then turn it like a clock.

Violation: Fold made before its turn

- **Visible cue:** a side is folded in while the collar still lies toward the front edge, or an end is folded while the strip still lies straight across the desk.

- **SOP rule broken:** Steps 8 and 10, the shirt is turned 90 degrees clockwise before the side folds, and 45 degrees clockwise before the end folds.
- **Coaching note:** turn, then fold. Each fold comes after its turn.

Violation: Folds made out of order

- **Visible cue:** the far side is folded in before the near side, or the far end is folded down before the near end is folded up.
- **SOP rule broken:** Steps 9 and 11, the near side is folded first, then the far side. The near end is folded up first, then the far end down.
- **Coaching note:** near first, then far, for the sides and for the ends.

Violation: Side fold edge off line

- **Visible cue:** a side fold edge runs crooked from the collar end to the hem, stops well short of or well past the end of the collar, or leaves a sleeve or cuff showing past the fold edge.
- **SOP rule broken:** Step 9, each fold edge runs straight and lines up with the end of the collar.
- **Coaching note:** line the fold up on the collar before you lay it down.

Violation: Fold made with one gripper

- **Visible cue:** a side fold or the end fold is made by one gripper while the other stays off the cloth.
- **SOP rule broken:** Steps 9 and 11, both grippers pinch the side or end and fold it together.
- **Coaching note:** two grippers, one at each end of the fold.

Violation: End fold off line

- **Visible cue:** the hem end covers much more or much less than the near third, the collar end is laid down well short of or well past the near fold edge, an end lies at an angle, or the collar end is folded under instead of on top.
- **SOP rule broken:** Step 11, the hem end covers about the near third of the strip, and the collar end is laid on top lined up with the near fold edge.
- **Coaching note:** thirds, hem first, collar on top and square.

Violation: Fold made by dragging

- **Visible cue:** a sleeve, a side, or an end is folded by pushing or sliding it across the desk instead of lifting it, carrying it over, and laying it down. A push to adjust a fold that was already lifted and laid down is an adjustment.
- **SOP rule broken:** Steps 7, 9, and 11, the gripper lifts the cloth before it carries it and lays it down.
- **Coaching note:** lift it to fold it. Push only to adjust.

Violation: Packet not turned or placed in the finished spot

- **Visible cue:** the packet is left in the work zone, laid down outside the finished spot, laid with the collar facing the front edge or a side, pushed or dragged to the spot, or let go above the desk so it drops.
- **SOP rule broken:** Step 12, the right gripper turns the packet until the collar faces the far side, carries it to the finished spot for the config, back-right or right-center, and lays it down there.
- **Coaching note:** turn it so the collar faces away, then carry it to the finished spot for the config.

Violation: Step skipped

- **Visible cue:** a step or substep is left out: the shirt is never spread before buttoning, a button is never worked, the placket is never smoothed, the shirt is never straightened after the flip, a sleeve is never folded onto the back, a turn is never made, a side or an end is never folded, or the episode moves on to a later step with an earlier one not done.
- **SOP rule broken:** Steps 1 to 12, every step and substep is run, in order.
- **Coaching note:** run every step, every time. If one was missed, go back and do it before moving on.

Violation: Wrong arm did the work

- **Visible cue:** the left gripper smooths the placket; a sleeve is brought to the middle of the back by the gripper on the other side, or the gripper that brought it also lays the rest of it; or the left gripper turns or carries the packet in Step 12. An adjustment by either gripper is not wrong arm.
- **SOP rule broken:** Steps 4, 7, and 12, the left gripper holds the middle of the hem, brings the left sleeve to the middle of the back, and finishes laying the right sleeve. The right gripper smooths the placket, brings the right sleeve to the middle of the back, finishes laying the left sleeve, and turns and places the packet.
- **Coaching note:** each arm keeps its own role and its own side.

Violation: Shirt dropped, off the work zone, or arms collided

- **Visible cue:** the shirt falls from a hold, a gripper opens above the desk so the cloth drops, part of the shirt ends up off the work zone or off the desk, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 12, keep the shirt in the work zone until Step 12 lays it in the finished spot, and let go only once the cloth is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a button open, a sleeve showing, a fold not made, the packet not in the finished spot, or an arm away from home; or an arm is sent home before the look at the packet.
- **SOP rule broken:** Step 13, look once at the packet, return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective button or buttonhole:** a button comes loose or off during the episode, or a buttonhole is too tight to pass its button on a correct draw. Replace the shirt before the next episode.
- **Defective shirt:** found torn, stained, damp, or with an open seam.
- **Shirt too large for the cell:** it cannot lie spread in the work zone with clear desk all round, or the hem still touches the desk when the shirt hangs from both shoulder points.

Annotation subtasks (from SOP)

1. Move the crumpled shirt from the shirt spot to the middle
2. Spread the shirt face up by the shoulders
3. Button one button
4. Smooth the placket
5. Flip the shirt face down by the shoulders
6. Straighten the shirt face down
7. Fold one sleeve onto the back
8. Turn the shirt 90 degrees clockwise
9. Fold one side in
10. Turn the strip 45 degrees clockwise
11. Fold one end over
12. Turn the packet and lay it in the finished spot
13. Return both arms home and end the episode